A Methodology for Transforming Natural Language Academic Policies into Formal Knowledge for Automated Reasoning

by

Ponkrit Kaewsawee

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Examination Committee:	Dr. Chaklam Silpasuwanchai (Chairperson)
	Dr. Chutiporn Anutariya
	Prof. Attaphongse Taparugssanagorn
Nationality:	Thai
Previous Degree:	Bachelor of Engineering in Electical Engineering
	Sripatum University, Bangkok
	Thailand

Scholarship Donor: Royal Thai Government

Asian Institute of Technology
School of Engineering and Technology
Thailand
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CHAPTER 1
INTRODUCTION

1.1 Background of the Study

Automated Compliance Checking (ACC) has emerged as a critical field for enforcing regulations across various domains, from legal tech to academic administration Anim et al. (2024). In academic institutions such as Asian Institute of Technology or AIT, policies and procedures (P&P) are typically stored as natural language documents. While easily readable by humans, these documents present a significant challenge for automated computer processing and reasoning. This manual reliance creates a risk of misinterpretation by stakeholders, especially students and prevents the institution from efficiently monitoring policy compliance.

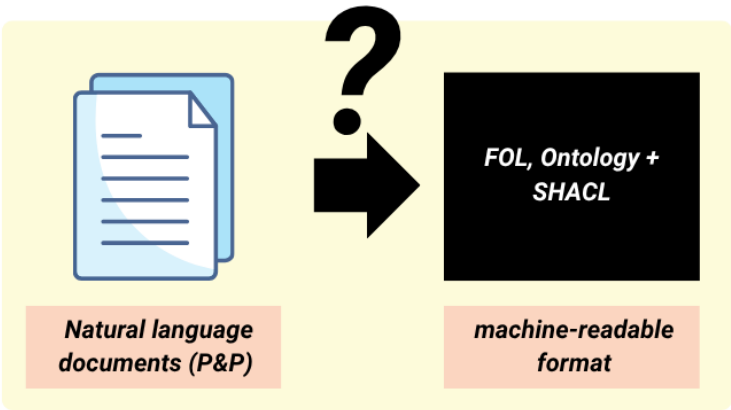
Recent studies have demonstrated the effectiveness of Semantic Web technologies for ACC, where an Ontology is used to formally model the concepts and relationships within a domain, and the Shapes Constraint Language (SHACL) is used to validate data against formal rules Anim et al. (2024). This approach moves towards standardized and interoperable solutions. Furthermore, theoretical research has established a strong formal connection between SHACL and First-Order Logic (FOL), enabling the use of FOL for static analysis and formally-grounded solutions for SHACL decision problems Pareti (2024). However, while these technologies for representation and validation are robust, a crucial gap remains between the original unstructured policies and their formal implementation.

1.2 Statement of the Problem

While the tools for validating data (SHACL) and the theoretical foundations (FOL) are well-studied, there remains a lack of a systematic methodology to transform natural language policies into a formal knowledge base. While frameworks like SHACL2FOL focus on translating existing SHACL rules *into* FOL for theoretical analysis Pareti (2024), there is no established, repeatable process for the initial and most critical step: translating unstructured text *through* a logical representation like FOL to a functional implementation with Ontology and SHACL. This absence of a structured bridge between human-readable text and machine-readable logic, as illustrated in Figure 1.1, is the core problem this research aims to solve.

Figure 1.1

The Methodological Gap in Transforming Policies into Formal Knowledge.



1.3 Research Questions

To address the stated problem, this research seeks to answer the following primary question:

RQ1: What constitutes a systematic methodology for transforming academic regulations from natural language policies into a formal knowledge base, utilizing First-Order Logic (FOL) as an intermediate representation and Semantic Web technologies (Ontology, SHACL) for implementation?

1.4 Hypotheses

The following hypotheses will guide this study:

H1: A systematic methodology using FOL will be able to formalize even complex, multi-conditional rules into a consistent, machine-verifiable format.

H2: The knowledge base created by this methodology will be sufficiently robust to serve as the foundation for a functional automated reasoning application, such as a proactive compliance checker.

1.5 Objectives of the Study

The main objectives of this study are:

1. To design and propose a systematic methodology for the analysis, formalization, and modeling of academic regulations from unstructured P&P documents.
2. To define a process that uses First-Order Logic (FOL) as an intermediate layer to capture the precise logic of a rule before its implementation in a specific technology, such as Ontology, SHACL.

3. To demonstrate the feasibility and effectiveness of the proposed methodology by developing a Proof-of-Concept (POC) knowledge base using selected AIT policies.

1.6 Methodology

This study proposes a systematic framework consisting of five main phases. The process begins with (1) **Classifying** AIT policies based on their complexity and clarity levels. Following this, the framework (2) **Defines** the appropriate FOL syntax for each policy class. The core of the methodology is (3) **Translating** the natural language policy statements into formal FOL statements; this phase incorporates principles from Assumption-Based Argumentation Dung et al. (2009) to analyze and resolve potential ambiguities before formalization. Subsequently, the framework details the process of (4) **Converting** the FOL components into an Ontology and SHACL rules (collectively termed a Shape Graph). Finally, the methodology is validated through (5) **Evaluation**, where student data (Data Graph) is checked against the developed Shape Graph using SHACL.

1.7 Scope

The scope of this research is defined as follows:

1. The initial policy classification is performed manually.
2. Due to time constraints, this research will initially focus only on policies rated as "Clear" according to the established definition.

1.8 Key Findings

As this is a proposal, this section outlines the expected results. The proposed methodology is expected to achieve high accuracy (defined as 100%) in detecting violations for low-complexity rules. For medium and high-complexity rules, an accuracy of approximately 50% is hypothesized. These results will establish a quantitative baseline for future research in this area.

1.9 Contributions

The primary contribution of this research is the proposal of a practical methodology. The key contributions are:

1. A framework and guideline to translate natural language policy statements into First-Order Logic (FOL) statements.
2. A framework and guideline to convert FOL statements into an implementation using Ontology and SHACL.
3. A demonstration, via a Proof-of-Concept, that this methodology is feasible for modeling and implementing a policy compliance checking system.

CHAPTER 2
RELATED WORK

This chapter reviews the existing literature relevant to the automatic verification of compliance in data-driven systems. The review is structured to build a case for this thesis by examining three key areas: (1) existing frameworks for data quality and validation; (2) the application of Semantic Web technologies, specifically OWL and SHACL, for formal validation; and (3) the use of Knowledge Graphs and Large Language Models within Explainable AI (XAI) to make system outputs more understandable. The primary goal of this chapter is to identify the research gap that our proposed framework aims to fill.

2.1 Data Quality and Validation Frameworks

The importance of maintaining high data quality, especially in pipelines integrating data from diverse sources, has led to the development of several validation frameworks. These frameworks aim to systematically detect and manage data defects. A prominent example is GraphGuard, a framework proposed by Dorsch et al. (2023) to improve data quality in Knowledge Graph (KG) pipelines.

GraphGuard introduces the concepts of "QualityContracts" and "Guardians" to formalize and automate the execution of validation rules. The framework defines a comprehensive ontology to represent validation rules, their execution process, and the resulting reports in a machine-readable format. This approach is powerful for managing validation within an existing data pipeline and aligns with our goal of systematization. However, the framework's starting point is the "QualityContract" itself; it presumes that these formal, machine-readable rules are already defined. It does not address the foundational challenge of how to systematically derive these contracts from high-level, unstructured source documents like institutional policies. This is a critical gap, as the difficulty often lies in the initial translation from human language to formal logic. Our work focuses precisely on this initial, crucial step that precedes the implementation of such a framework.

2.2 Automated Compliance Checking with Semantic Web Technologies

The field of Automated Compliance Checking (ACC) has gained significant attention as a means to enforce regulations in complex domains. A leading-edge approach in this area involves the use of Semantic Web technologies. A prominent example is the work by Anim et al. (2024), which presents a SHACL-based approach for ACC in the legal domain. Their research successfully demonstrates the power of SHACL-SPARQL to handle regulations with nuanced temporal and aggregate requirements, such as those found in Ghana's petroleum industry. Their case study proves that these technologies are not only feasible but also effective for real-world, complex compliance scenarios.

However, the primary focus of their work is on the execution and validation of rules that are already formalized within the Semantic Web environment. The paper acknowledges but does not address the preceding, and arguably more challenging, task: how to systematically derive these formal SHACL rules from the original, unstructured legal texts. This leaves a significant methodological gap, which is the starting point for our research.

2.3 Theoretical Foundations: Linking SHACL and First-Order Logic

To create a robust and verifiable transformation process, it is essential to ground it in formal logic. First-Order Logic (FOL) serves as the universally understood language of formal semantics. Recent theoretical work has focused on creating a solid bridge between the practical SHACL standard and FOL. The work of Pareti (2024) is central to this effort, introducing SHACL2FOL, a toolkit designed to automatically translate SHACL documents into FOL sentences. This translation enables the formal analysis of SHACL shapes, allowing for crucial checks like satisfiability (Can any data ever conform to this rule?) and containment (Is this rule a stricter version of another rule?).

Pareti's work provides a vital theoretical validation for the link between SHACL and FOL. However, its operational direction is from SHACL to FOL, designed for the analysis of *already-existing formal rules*. It does not offer a methodology for the reverse—and arguably more practical—problem of formalizing unstructured natural language *into* a logical representation like FOL, which can then be implemented as SHACL. Our research addresses this opposite direction, focusing on creation rather than post-analysis.

2.4 Formal Argumentation for Rule Analysis

When translating natural language, ambiguity and potential conflicts are inevitable. Formal argumentation theories provide a robust way to handle such issues. Dung et al. (2009) introduced Assumption-Based Argumentation (ABA), a computational framework for default reasoning. ABA is inspired by how humans reason with incomplete information and resolve conflicts by evaluating the "acceptability" of arguments based on the assumptions they rest upon. It provides a powerful, abstract framework for modeling dialectical reasoning, where arguments and counter-arguments are formally structured and evaluated.

The main limitation of ABA, for practical purposes, is its highly theoretical nature. The original work does not offer a concrete methodology for applying these abstract principles to a specific, real-world domain like formalizing academic policies. It provides the "what" (a theory of acceptable arguments) but not the "how" (a step-by-step process to model a specific problem). Our work aims to make this theory practical by integrating the principle of analyzing arguments into our knowledge modeling process, using Semantic Web tools as the concrete implementation technology.

2.5 Synthesis and Research Gap

The literature review reveals a clear and compelling research gap, as summarized in Table 2.1. On one hand, the field of ACC has powerful implementation technologies like SHACL Anim et al. (2024). On the other hand, robust theoretical foundations exist in both formal logic Pareti (2024) and argumentation theory Dung et al. (2009). However, no existing work provides a comprehensive, systematic methodology that connects these pieces. There is no established framework to guide a researcher or practitioner through the entire process: from an unstructured policy document, through a rigorous logical formalization, to a final, verifiable, and machine-readable knowledge base.

Table 2.1

Positioning the Proposed Methodology against Key Related Work.

Research Paper	Primary Focus	Key Contribution	Limitation / Gap Addressed by Our Work
Anim et al. (2024)	Execution of formal rules for Automated Compliance Checking (ACC).	Demonstrates the power of SHACL-SPARQL for complex, real-world temporal and aggregate compliance rules.	Focuses on the execution of rules, but does not propose a systematic methodology for creating these formal rules from natural language.
Pareti (2024)	Theoretical analysis of SHACL rules.	Provides a toolkit (SHACL2FOL) to translate SHACL into FOL, enabling formal proofs of satisfiability and containment.	The translation direction is SHACL $\rightarrow$ FOL for analysis. It does not address the practical process of converting natural language $\rightarrow$ FOL $\rightarrow$ SHACL for implementation.
Dung et al. (2009)	Abstract framework for default reasoning and conflict resolution.	Establishes Assumption-Based Argumentation (ABA) as a foundational theory for modeling arguments and determining their acceptability.	Highly theoretical and abstract. It does not offer a concrete methodology or technical tools to apply these principles to real-world modeling tasks.
This Thesis	A systematic methodology for transforming natural language policies into formal knowledge.	Proposes a complete, step-by-step framework that bridges the gap from unstructured text to an implementable knowledge base, using FOL as an intermediate layer.	Directly addresses the lack of a systematic process, making abstract theories (like ABA) practical and creating the formal knowledge that systems (like Anim et al.'s) require.

CHAPTER 3
Methodology

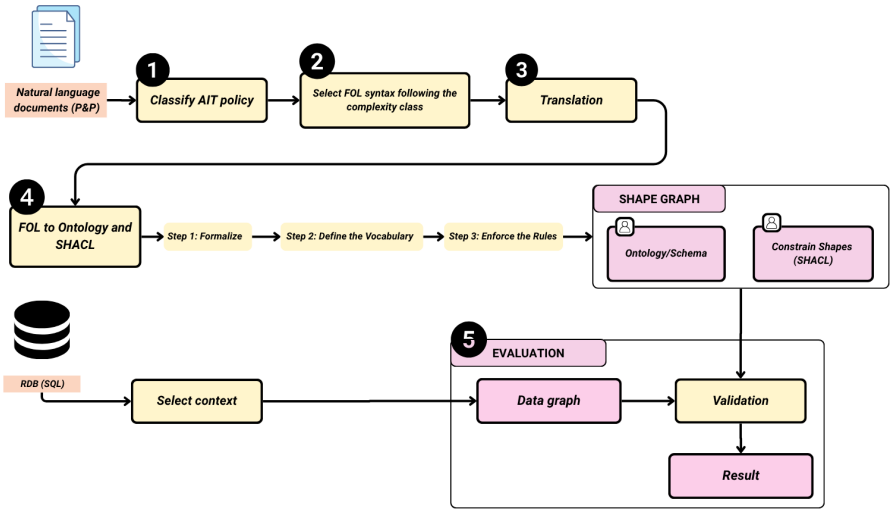
This chapter details the systematic methodology proposed for transforming natural language academic policies into a formal, machine-readable knowledge base for automated reasoning. This framework represents the core contribution of the thesis, providing a structured, repeatable process from unstructured text to a verifiable implementation. The methodology is designed in distinct phases, ensuring clarity and modularity at each step of the transformation process.

3.1 The Proposed Systematic Framework

The proposed methodology is organized into a systematic framework comprising five primary phases, as illustrated in Figure 3.1. The framework is designed to logically progress from high-level policy analysis to a concrete, testable knowledge base. It begins with the manual classification of policies, followed by their formalization into an intermediate logical representation, and concludes with their conversion into Semantic Web technologies for automated validation. The subsequent sections of this chapter will provide a detailed explanation of each phase.

Figure 3.1

The Proposed Systematic Framework for Transforming Policies into Formal Knowledge.



3.2 Dataset

The primary case study for this research utilizes official documents from the Asian Institute of Technology (AIT). The dataset consists of five key Policies and Procedures (P&P) documents:

- AA-4-1-1 Academic Integrity in Research and Publication
- FB-6-1-1 Credit Policy
- FS-1-1-1 Campus Accommodation for Students
- PA-2-1-2 Ethical Behavior and Grievance Process
- Student-Handbook

From these documents, the research scope is focused on a specific subset of regulations. To ensure feasibility, we will begin with rules that relate to the status of current students and are rated as "Clear" according to the definition provided in the next section. This selection process resulted in a working set of 38 distinct rules.

3.3 Phase 1: Policy Analysis and Classification

The first phase of the methodology involves a manual analysis and classification of the selected AIT policies. Each rule is rated along two key dimensions: **Complexity** and **Clarity**. This classification allows for a structured approach, beginning with simpler rules to establish a baseline before tackling more ambiguous or complex cases.

1. **Complexity Level:** This dimension assesses the logical structure of the rule.
    - **Low:** Simple, direct commands or prohibitions with no extra conditions (e.g., "No pets allowed.").
    - **Medium:** Rules containing 1-2 conditions, exceptions, or a short sequence of steps (e.g., "Cooking is not allowed, except in designated dorms.").
    - **High:** Rules involving multiple steps, dependencies on other departments (e.g., finance, visa), or complex penalty structures.
2. **Clarity Assessment:** This dimension assesses the ambiguity of the language used.
    - **Clear:** The statement requires no interpretation and can be followed immediately.
    - **Partially Ambiguous:** The statement is mostly clear but contains vague terms that could lead to different interpretations (e.g., the term "large electrical appliances" is not defined).
    - **Ambiguous:** The statement is unclear, and its true intention cannot be understood without additional information.

3.4 Phase 2: Formalization into First-Order Logic (FOL)

This phase focuses on translating the classified policy statements into a formal, intermediate representation using First-Order Logic (FOL). FOL is chosen for its expressive power and its ability to capture the precise logical structure of a rule, independent of any specific implementation technology. The translation process involves:

1. **Simplifying the Policy Statement:** The original text is rephrased into a clear, logical sentence.
2. **Extracting Vocabulary:** Key concepts are identified and categorized as **Variables** (e.g., Student(x)), **Functional Predicates** (attributes, e.g., StudentType(x, t)), and **Relational Predicates** (relationships, e.g., HasObligation(x, y)).
3. **Analyzing Arguments:** Principles from Assumption-Based Argumentation (Dung, 1995, 2009) are used at a conceptual level to analyze the logical structure, identify underlying assumptions, and resolve potential conflicts within the rule's conditions.
4. **Constructing the FOL Statement:** The extracted vocabulary and logical structure are combined to form a complete FOL statement, typically following an "If...Then..." (implication) structure.

3.5 Phase 3: Conversion from FOL to Ontology and SHACL

This phase provides a practical pattern for converting the abstract FOL statements into a concrete implementation using Semantic Web technologies. This process is broken down into three steps:

1. **Step 1: Formalize:** This initial step, covered in the previous phase, ensures a clear FOL statement with an extracted vocabulary is ready for conversion.
2. **Step 2: Define the Vocabulary (Ontology Creation):** The vocabulary extracted during the FOL analysis is formally defined in an Ontology. This process is analogous to building a data dictionary for the knowledge base.
   - Every concept (e.g., Student) is mapped to an `rdfs:Class`.
   - Every relation is mapped to an `rdf:Property`. This includes creating Object Properties for linking two classes (e.g., `hasFinancialObligation` links Student to Obligation) and Data Properties for linking a class to a literal value (e.g., `hasObligationType`).
3. **Step 3: Enforce the Rules (SHACL Shape Creation):** The propositional structure of the FOL statement (If...Then...) is converted into a **SHACL Shape**. This

shape defines the constraints for validating the data.

- The universal quantifier ("For all students x...") defines the `sh:targetClass` of the shape (e.g., `ait:Student`).
- The "If" part of the FOL statement becomes the conditions checked within the SHACL constraint, often implemented as a `sh:sparql` constraint to handle complex patterns.
- The "Then" part of the FOL statement defines the outcome of the constraint. If the "If" conditions are met, it signifies a potential violation, which is then reported by the SHACL validator.

3.6 Phases 4 & 5: Implementation and Evaluation

The final phases involve implementing the knowledge base and evaluating its effectiveness. This is achieved by creating two key components derived from the previous steps:

1. **The Shape Graph:** This is the formal representation of the academic policies, consisting of the Ontology (the vocabulary) and the SHACL Shapes (the rules).
2. **The Data Graph:** This contains the actual data to be validated. For this research, anonymized student data is fetched from a relational database and converted into an RDF graph format that conforms to the structure of the Ontology.

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